# Imaging Methods (PSYC 160) Final Exam Writeup - Gabe Fajardo

I ran the analyses primarily on the HPC. To optimize the workflow and set up future analyses on larger datasets, I wrote different scripts for each analysis. For convenience, I've included PDFs of the marimo python scripts and their outputs in this Google Drive folder, and I also include main output images in this writeup as well. The full scripts can also be found on the HPC at this path: `/dartfs/rc/lab/D/DBIC/DBIC/psych160/students/gabef/exam`

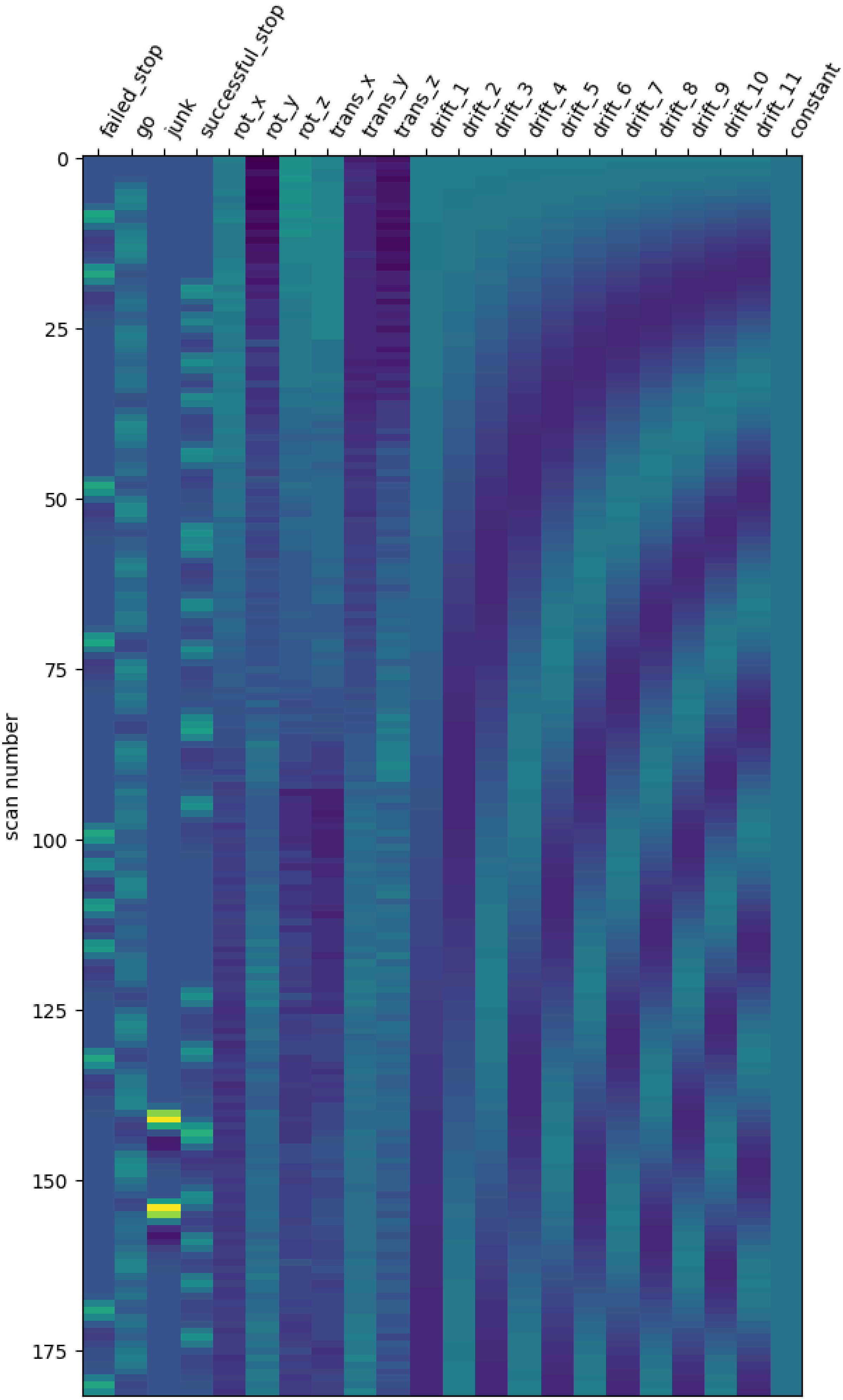
Note: these scripts were initially ran in my lab's shared space, but I copied them to the course directory in case there are any access restrictions.

## 1. Preprocessing + univariate GLM

I first ran `fMRIPrep` on subs 01 and 02. The bash script can be found in the `preproc` folder. PDFs of the HTML QC output can be found in the Google Drive folder (sub-01 and sub-02).

For the 1st level GLM, I ran this individually for each subject using `Nilearn`. I first gathered key hyperparameters from the paper: TR=2s, smoothing FWHM=5, and a high-pass filter = 1/66. I tried to figure out which specific nuisance regressors they used in their design matrix, but this was a bit unclear. I chose to include the 6 basic motion confounds and the default cosine drift model implemented in the `FirstLevelModel` function. To get the events df in the right format, I first had to first remove the first row (duration=NaN) and remove spaces from the trial_type condition names. [FULL SCRIPT]

Here is a plot of an example design matrix:

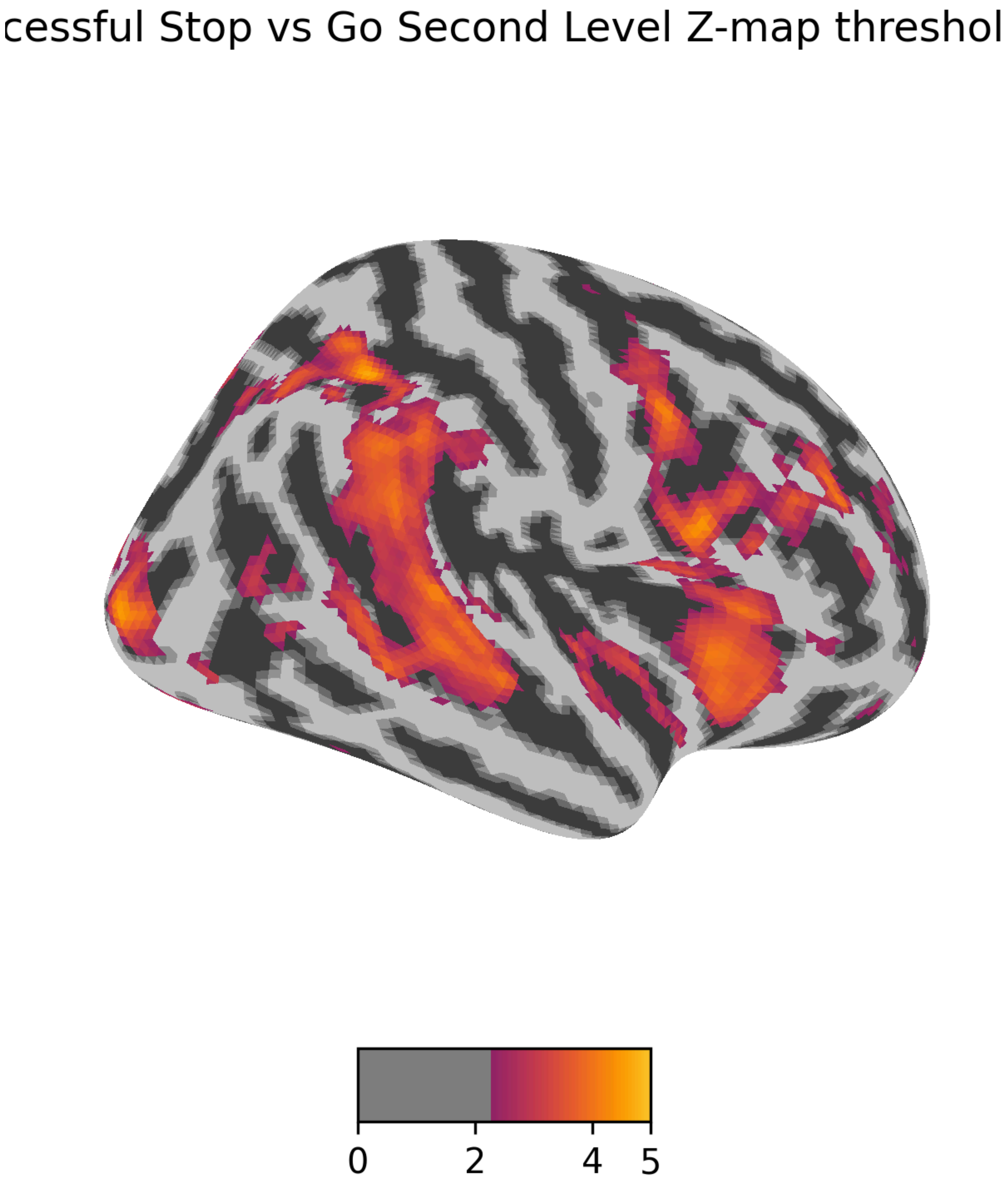

failed_stop
go
junk
successful_stop
rot_x
rot_y
rot_z
trans_x
trans_y
trans_z
drift_1
drift_2
drift_3
drift_4
drift_5
drift_6
drift_7
drift_8
drift_9
drift_10
drift_11
constant
scan number
0
25
50
75
100
125
150
175

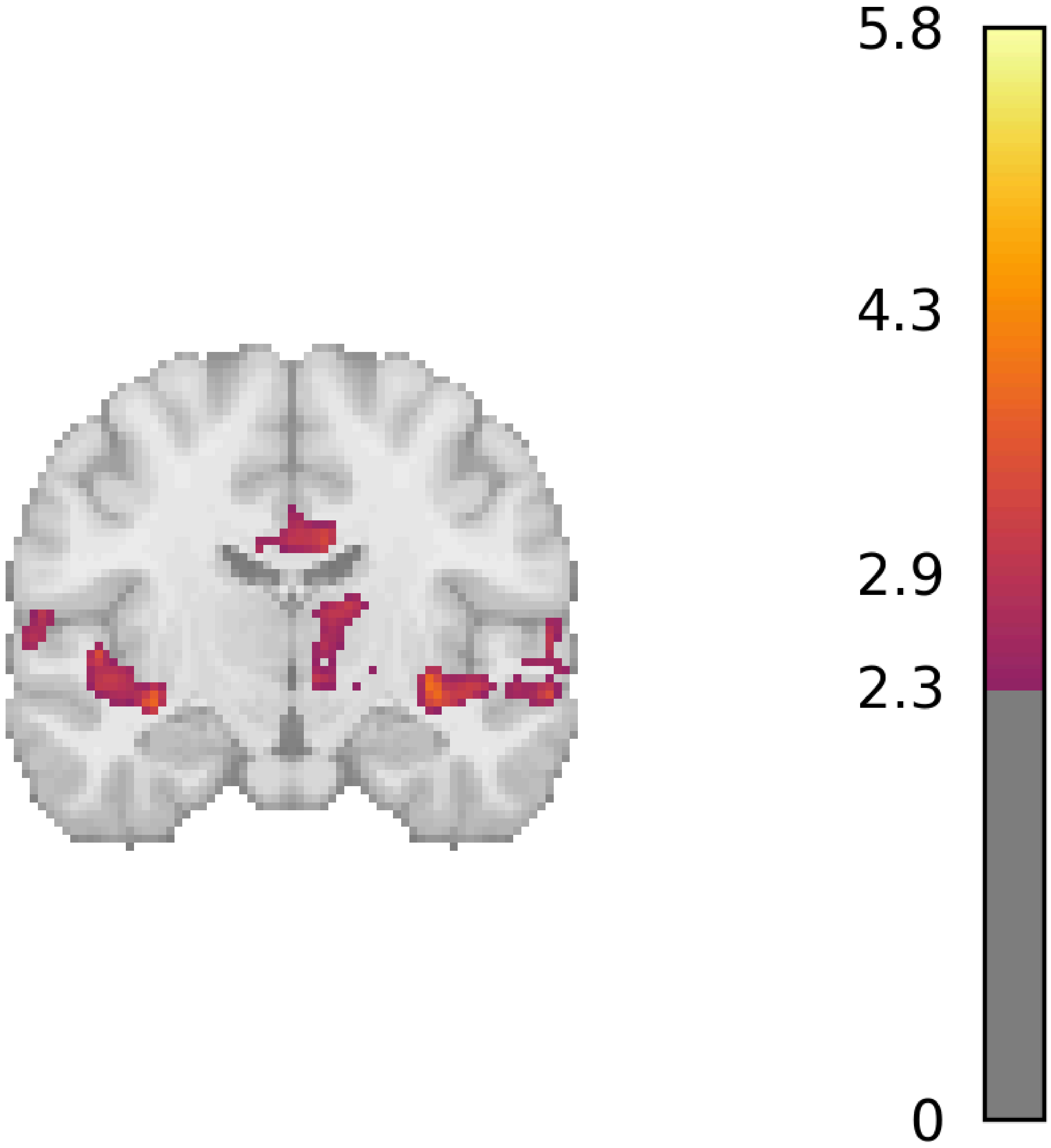
I then loaded the events, fMRI BOLD images, and confounds, and fit the GLM. Note, run-03 was missing for sub-12, so I excluded that run and ran the analysis for sub-12 on the remaining 2 runs.

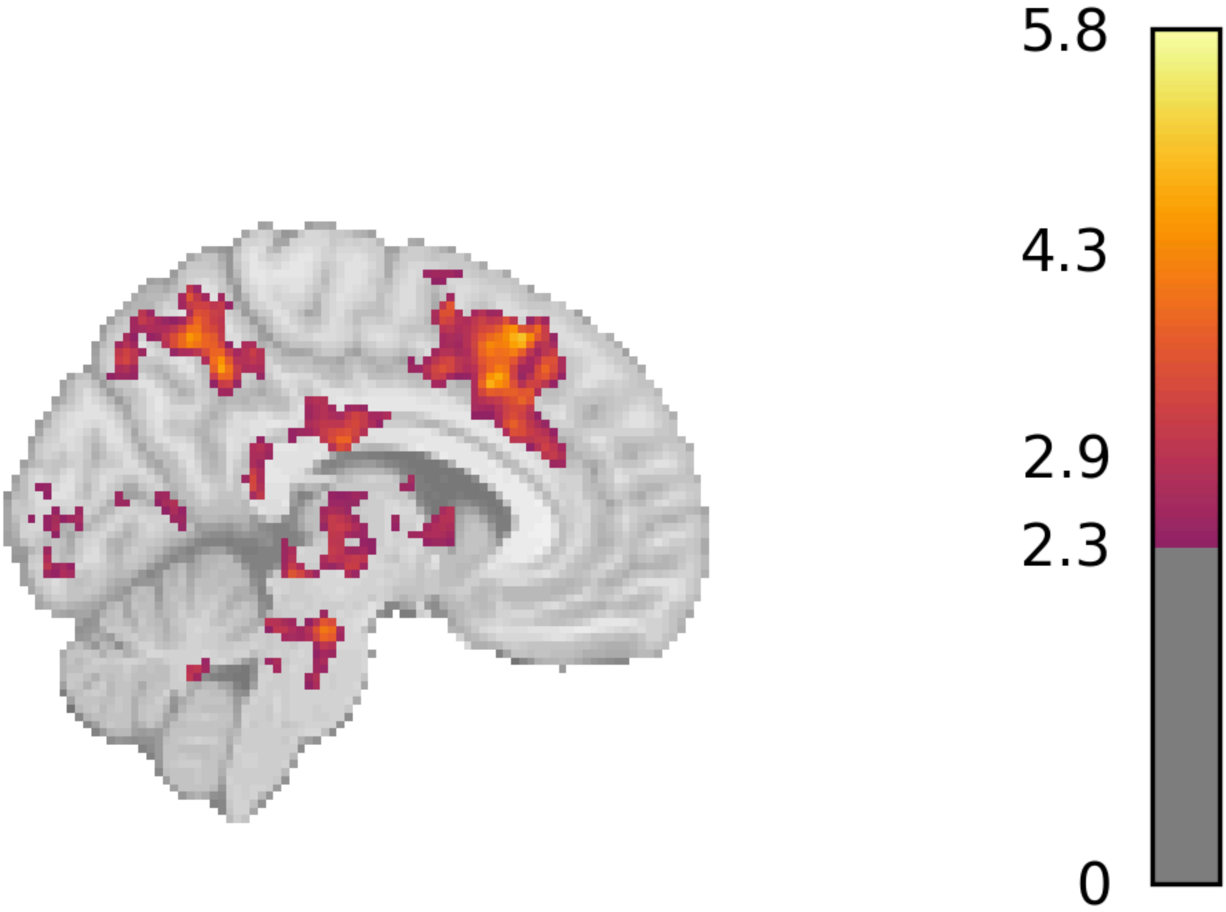
For each subject, I computed the desired contrast of `successful_stop > go` (`output="effect_size"` to get betas), and saved the beta map. I then took these beta maps, 1 for each subject, and used the `SecondLevelModel` function, where I estimated the group-level effect via the `intercept` contrast, yielding a group-level Z-map.

The cluster-wise thresholding they used (GRFT) was not readily available in `Nilearn`, so I instead applied a voxel-wise threshold of `z>2.3` and set an arbitrary cluster size of `> 30 voxels`. I plotted both surface plots and stat_map plots, similar to the paper. All code can be seen in the exported PDF of the 2nd level marimo notebook. [FULL SCRIPT]

Here are some plots:


cessful Stop vs Go Second Level Z-map threshol
0
2
4
5


5.8
4.3
2.9
2.3
0

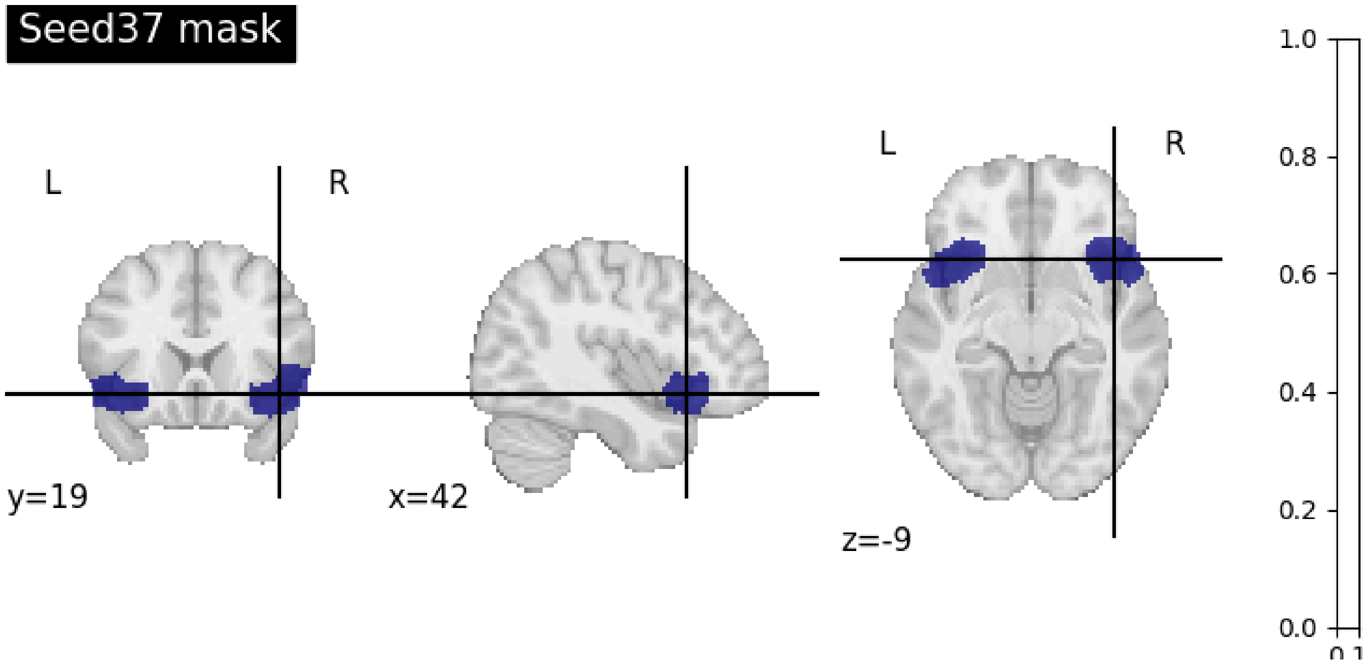


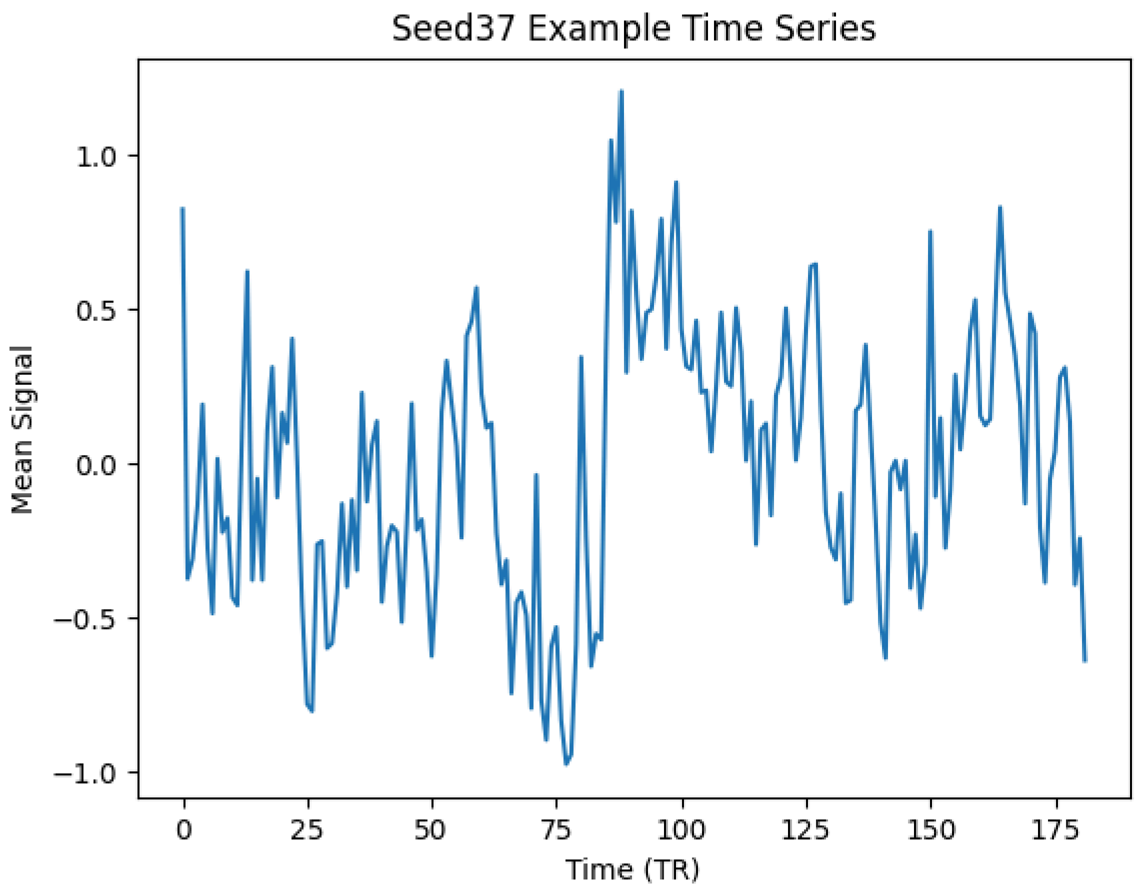

I found similar results as reported in the paper (fig. 2c): significant activation for `succesful stops` (greater than `go`) in the right STN, right pre-SMA, IFC, right Globus Pallidus (GP), right parietal cortex, and right insula.

## 2. Connectivity

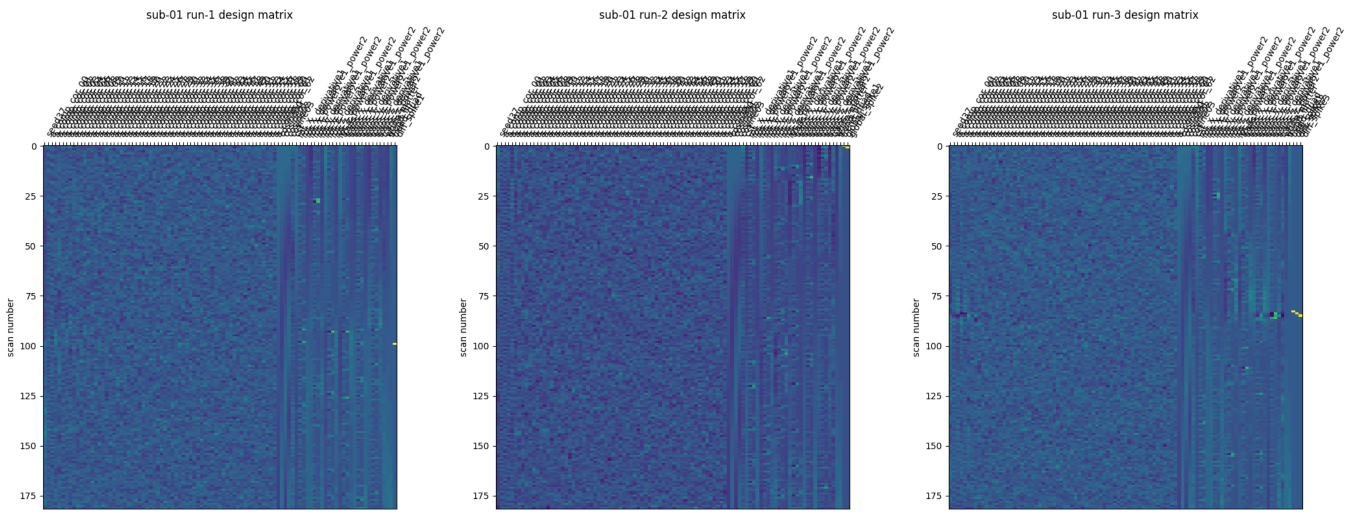
I first ran the connectivity analysis for each subject using the GLM-based approach discussed in class. I smoothed the fMRI data (FWHM=6), and then used the `load_confounds` function to load the appropriate nuisance regressors: full motion (24 params), wm + csf, and compcor components. I also included linear and quadriatic drift trends, as well as global and local spike regressors.

I then extracted and z-scored the timeseries for the anterior insula mask (seed37 from the k50 parcellation mask). This time series and all the nuisance regressors mentioned above were combined into a design matrix for each run. We got the seed37 contrast (controlling for the covariates) for each subject, giving us which voxels coactivate with the signal in the anterior insula. [FULL SCRIPT]



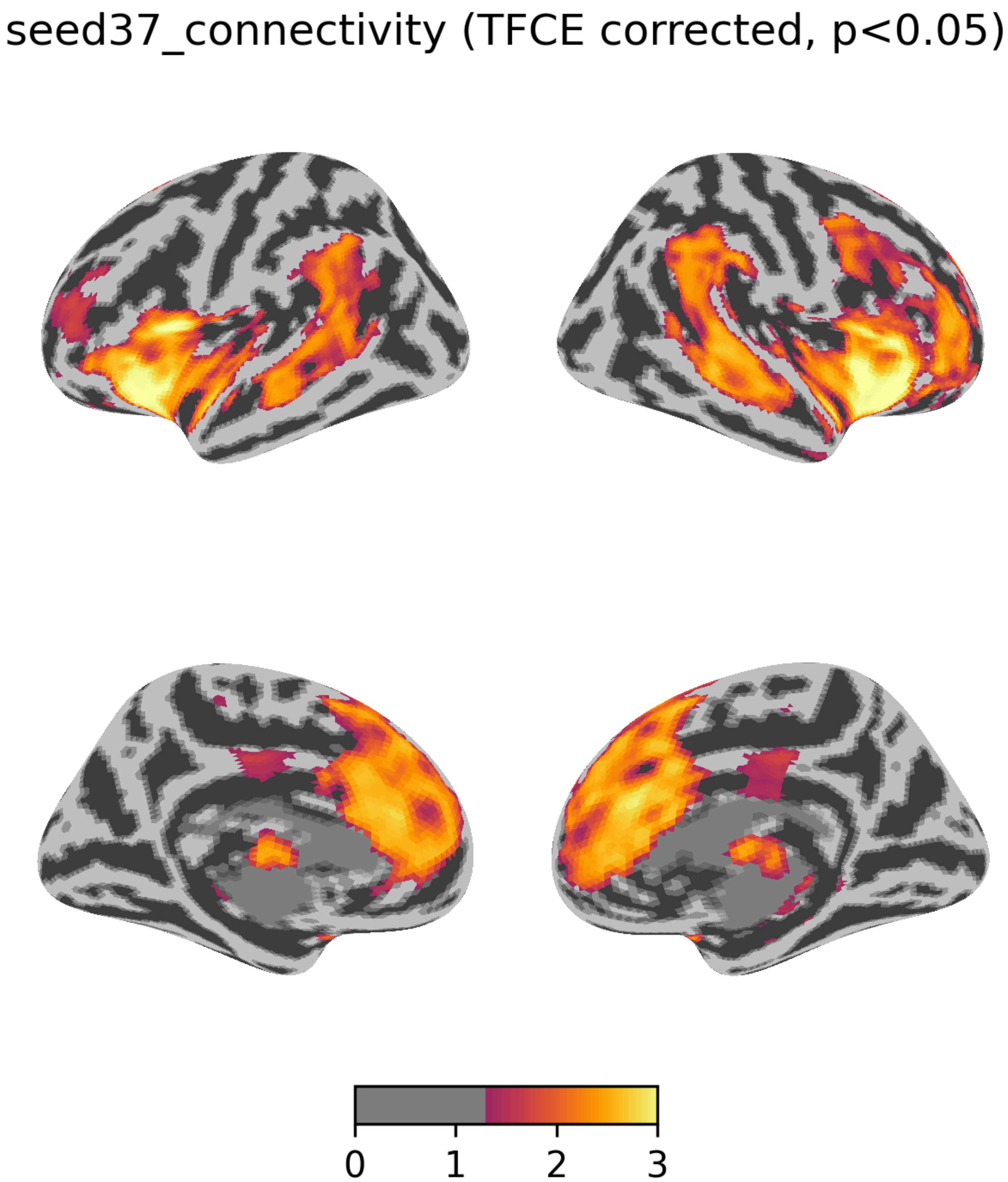




Examples design matrices:




I then ran the a 2nd level model on these beta maps to get a group-level zmap. This map had lots of activation at different thresholds, so I decided to also run a stricter TFCE permutation correction. [FULL SCRIPT]

Here is the resulting surface plot:

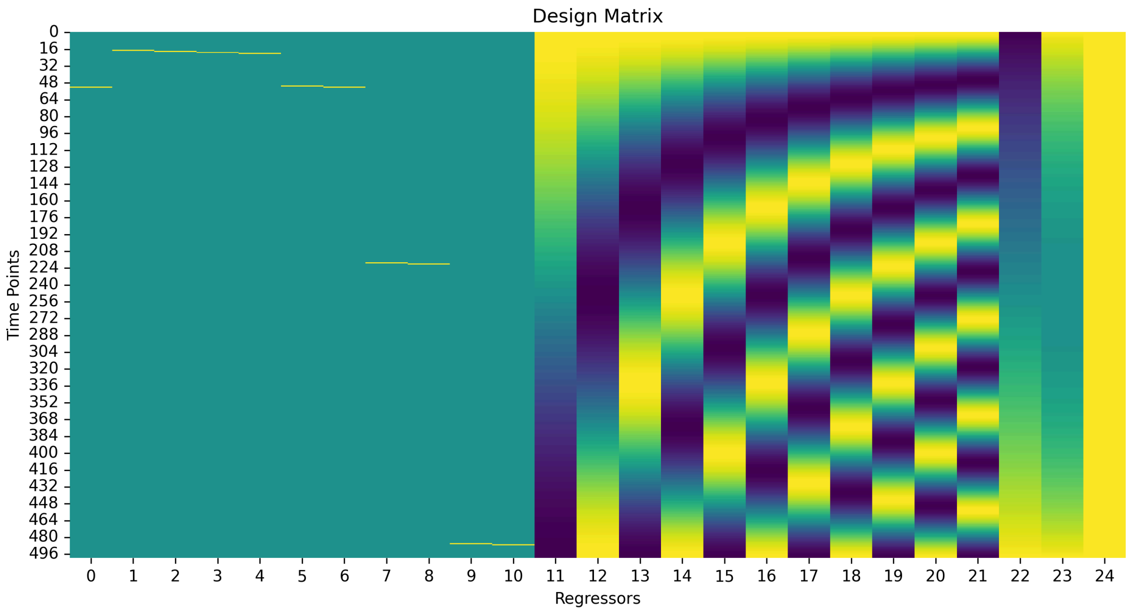



As expected, we find significant coactivation in the bilateral anterior insula. We also see coactivation in the bilateral STS ant TPJ, as well as the mPFC, cingulate cortex, caudate nucleus, and thalamus.

## 3. Denoise

I used global and local spikes, discrete cosine transforms, and polynomial drift nuisance variables to regress out the noise in the signal. The denoised signal can be found in `Denoised_Signal.csv`. [FULL

SCRIPT]




From the plot below, we can clearly see that spikes were removed. We see the clear downwards trend removed as well.